

Matplotlib Assignment 5

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1. Draw line chart to show urban and rural population of India over the years using population.csv [rural population is the difference between total population and urban population]

```
In [70]: import pandas as pd  
import numpy as np  
from matplotlib import pyplot as plt
```

```
In [71]: df = pd.read_csv('Population.csv')  
df
```

Out[71]:

	Year	Population	Yearly % Change	Yearly Change	Median Age	Fertility Rate	Density	Urban Pop %	Urb Populati
0	1955	3.877009e+08	2.29%	8284413.0	19.7	5.91	130	18.60%	7195849
1	1960	4.359903e+08	2.38%	9657890.0	19.2	5.92	147	18.50%	8056572
2	1965	4.901401e+08	2.37%	10829962.0	18.5	5.94	165	19.10%	9349384
3	1970	5.458643e+08	2.18%	11144824.0	18.1	5.62	184	20.00%	10938895
4	1975	6.113095e+08	2.29%	13089053.0	18.4	5.20	206	21.70%	13253381
5	1980	6.873540e+08	2.37%	15208898.0	18.9	4.78	231	23.40%	16094194
6	1985	7.726478e+08	2.37%	17058754.0	19.3	4.43	260	24.60%	19032178
7	1990	8.649722e+08	2.28%	18464886.0	19.7	4.04	291	25.70%	22229672
8	1995	9.603010e+08	2.11%	19065765.0	20.3	3.65	323	26.60%	25555882
9	2000	1.057923e+09	1.96%	19524338.0	21.2	3.35	356	27.50%	29135028
10	2005	1.154676e+09	1.77%	19350718.0	22.2	2.96	388	29.00%	33447940
11	2010	1.243482e+09	1.49%	17761048.0	23.6	2.60	418	30.60%	38074455
12	2015	1.328024e+09	1.32%	16908587.0	25.3	2.29	447	32.30%	42906945
13	2020	1.402618e+09	1.10%	14918639.0	27.0	2.05	472	34.40%	48309864
14	2022	1.425423e+09	0.81%	11402759.0	27.7	1.99	479	35.50%	50630486
15	2023	1.438070e+09	0.89%	12646384.0	28.1	1.98	484	36.00%	51823912
16	2024	1.450936e+09	0.89%	12866195.0	28.4	1.96	488	36.60%	53038714
17	2025	1.463866e+09	0.89%	12929734.0	28.8	1.94	492	37.10%	54274253



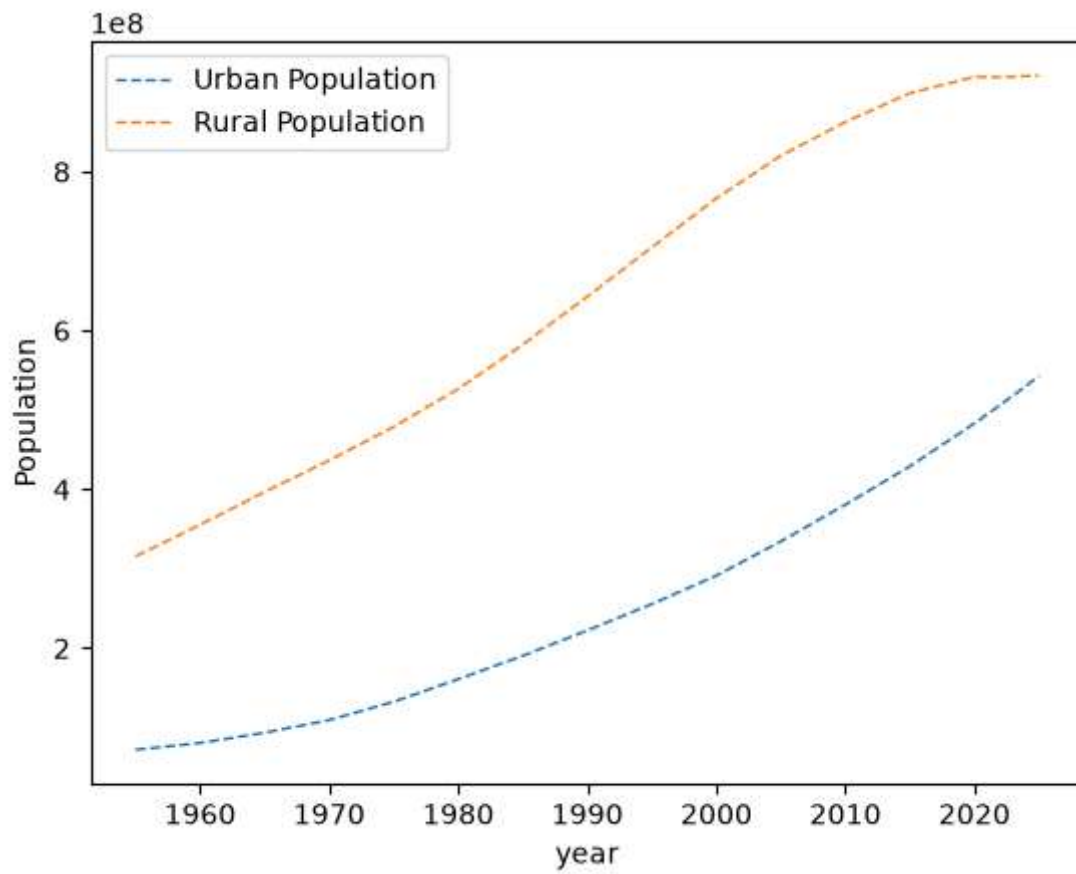
```
In [72]: df['Rural Population'] = df['Population'] - df['Urban Population']
```

```
In [73]: df['Year'].max()
```

```
Out[73]: np.int64(2025)
```

```
In [74]: plt.plot(df['Year'],df['Urban Population'],ls = '--' ,lw = 1)
plt.plot(df['Year'],df['Rural Population'],ls = '--' ,lw = 1)
plt.ylabel('Population')
plt.xlabel('year')
plt.legend(['Urban Population','Rural Population'])
plt.figure(figsize=(2,2 ))
plt.show()

# plots urban population vs rural population over year using dashed lines
```



<Figure size 200x200 with 0 Axes>

In [75]: df

Out[75]:

	Year	Population	Yearly % Change	Yearly Change	Median Age	Fertility Rate	Density	Urban Pop %	Urb Populati
0	1955	3.877009e+08	2.29%	8284413.0	19.7	5.91	130	18.60%	7195849
1	1960	4.359903e+08	2.38%	9657890.0	19.2	5.92	147	18.50%	8056572
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17	2025	1.463866e+09	0.89%	12929734.0	28.8	1.94	492	37.10%	54274253

2. Using a Bar chart display India's total, urban and rural population from year 2020 to 2025

In [76]: `df['Year'].loc[13:]`

Out[76]:

```

13    2020
14    2022
15    2023
16    2024
17    2025
Name: Year, dtype: int64

```

In [77]:

```

filtered_data = df[ (df['Year'] > 2019) & (df['Year'] < 2026 ) ]
filtered_data
# filter out the dataframe for years between 2019 and 2026

```

Out[77]:

	Year	Population	Yearly % Change	Yearly Change	Median Age	Fertility Rate	Density	Urban Pop %	Urb Populati
13	2020	1.402618e+09	1.10%	14918639.0	27.0	2.05	472	34.40%	48309864
14	2022	1.425423e+09	0.81%	11402759.0	27.7	1.99	479	35.50%	50630486
15	2023	1.438070e+09	0.89%	12646384.0	28.1	1.98	484	36.00%	51823912
16	2024	1.450936e+09	0.89%	12866195.0	28.4	1.96	488	36.60%	53038714
17	2025	1.463866e+09	0.89%	12929734.0	28.8	1.94	492	37.10%	54274253

```

In [78]: xwidth = 0.3
x_year = filtered_data['Year']
plt.figure(figsize=(12,6))
b1 = plt.bar(x_year-xwidth, filtered_data['Rural Population'],width =xwidth, label='Rural Population')
b2 = plt.bar(x_year, filtered_data['Urban Population'],width =xwidth, label='Urban Population')
b3 = plt.bar(x_year+xwidth, filtered_data['Population'],width =xwidth, label='Total Population')

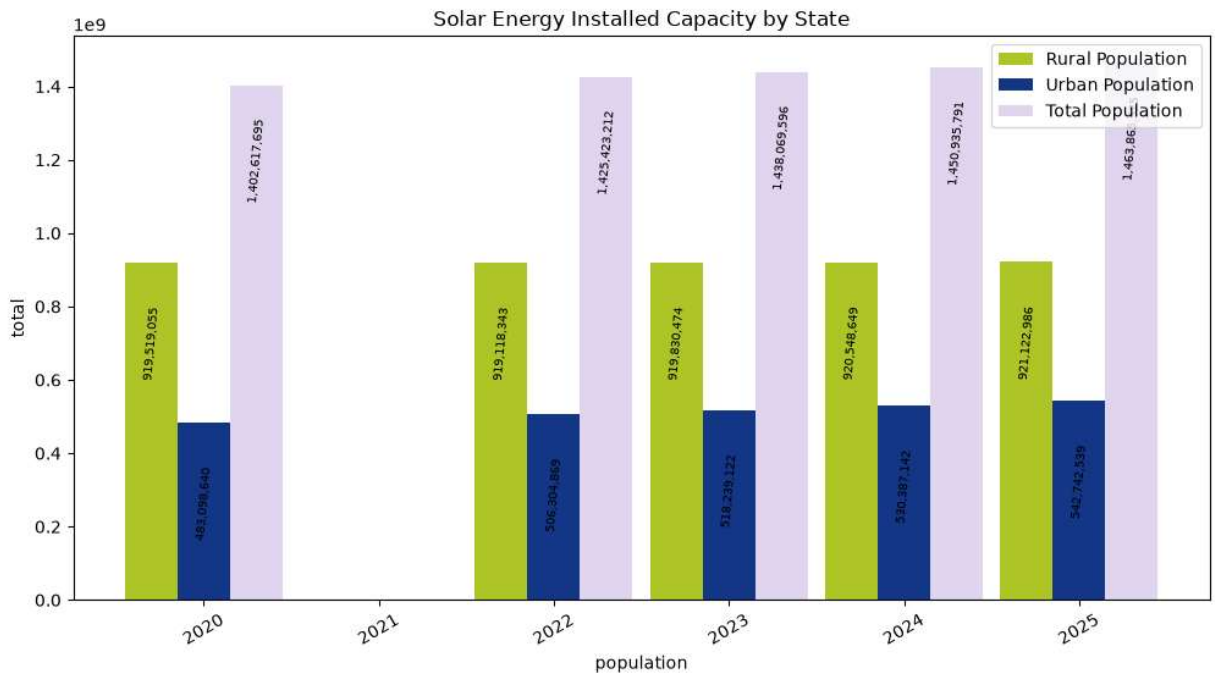
plt.bar_label(b1, fmt='{:, .0f}', padding=-70, rotation=85, fontsize=7)
plt.bar_label(b2, fmt='{:, .0f}', padding=-70, rotation=85, fontsize=7)
plt.bar_label(b3, fmt='{:, .0f}', padding=-70, rotation=85, fontsize=7)

plt.xticks(rotation=30)
plt.xlabel("population")
plt.ylabel("total")
plt.title("Solar Energy Installed Capacity by State")
plt.legend()

plt.show()

# this graph shows Rural, Urban, and Total Population side-by-side.
# stored x data means df[year] series of filtered data
# set size of figure for plotting
# created group bar chart using Rural Population , Urban Population and total Pop
# give the labels to each sub bars with their their values
# then rotated xticks with 30 degree angle
# set xlable, ylable , title, legend, and last show method

```

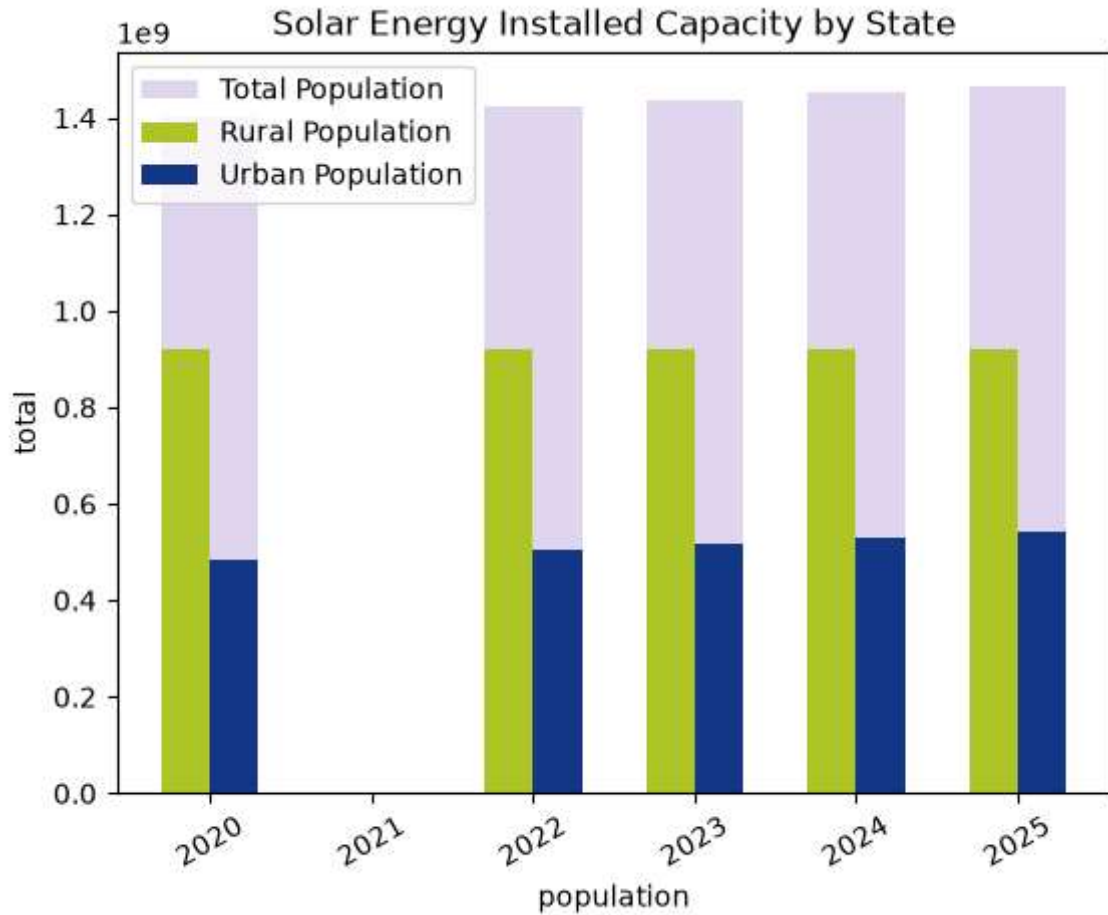


```
In [79]: xwidth = 0.3
x_year = df['Year'].loc[13:]

plt.bar(x_year, df['Population'].loc[13:],width =0.6, label='Total Population',color='green')
plt.bar(x_year-xwidth/2, df['Rural Population'].loc[13:],width =xwidth, label='Rural Population',color='blue')
plt.bar(x_year+xwidth/2, df['Urban Population'].loc[13:],width =xwidth, label='Urban Population',color='orange')

plt.xticks(rotation=30)
plt.xlabel("population")
plt.ylabel("total")
plt.title("Solar Energy Installed Capacity by State")
plt.legend()

plt.show()
```



In [80]: `df.dtypes`

```
Out[80]: Year          int64
Population    float64
Yearly % Change    str
Yearly Change    float64
Median Age      float64
Fertility Rate   float64
Density         int64
Urban Pop %      str
Urban Population float64
Country's Share  str
World Population float64
Rural Population float64
dtype: object
```

3. Using `gapminder2007.csv` show life expectancy of top 10 highly populated countries from Asia

```
In [81]: df = pd.read_csv('gapminder2007.csv')
df
```

Out[81]:

	country	pop	continent	lifeExp	gdpPercap
0	Afghanistan	31889923	Asia	43.828	974.580338
1	Albania	3600523	Europe	76.423	5937.029526
2	Algeria	33333216	Africa	72.301	6223.367465
3	Angola	12420476	Africa	42.731	4797.231267
4	Argentina	40301927	Americas	75.320	12779.379640
...	...	...	...	...	...
137	Vietnam	85262356	Asia	74.249	2441.576404
138	West Bank and Gaza	4018332	Asia	73.422	3025.349798
139	Yemen, Rep.	22211743	Asia	62.698	2280.769906
140	Zambia	11746035	Africa	42.384	1271.211593
141	Zimbabwe	12311143	Africa	43.487	469.709298

142 rows × 5 columns

```
In [82]: df.groupby('continent').get_group('Asia').sort_values('lifeExp',ascending=False).head(10)

# group the dataframe with continent then get asia group
# after this sorted values based on lifeExp column in descending order
# and then shown head(10) shows top 10 data
```

Out[82]:

	country	pop	continent	lifeExp	gdpPercap
66	Japan	127467972	Asia	82.603	31656.068060
55	Hong Kong, China	6980412	Asia	82.208	39724.978670
63	Israel	6426679	Asia	80.745	25523.277100
113	Singapore	4553009	Asia	79.972	47143.179640
70	Korea, Rep.	49044790	Asia	78.623	23348.139730
125	Taiwan	23174294	Asia	78.400	28718.276840
71	Kuwait	2505559	Asia	77.588	47306.989780
96	Oman	3204897	Asia	75.640	22316.192870
7	Bahrain	708573	Asia	75.635	29796.048340
137	Vietnam	85262356	Asia	74.249	2441.576404

In []:

3. Using a pie chart show distribution of top six selling cars [Create data as required]

```
In [83]: import random

random.seed(42)
car_names = [
    'Nexon', 'WagonR', 'Swift', 'Baleno', 'Brezza',
    'Punch', 'Creta', 'Venue', 'i20', 'Seltos',
    'Sonet', 'Thar', 'Scorpio', 'XUV700', 'Alto',
    'Ertiga', 'Fronx', 'Dzire', 'Harrier', 'Tiago'
]
data = {
    'name': car_names,
    'price': [random.randint(15000, 50000) for _ in car_names],
    'sell count': [random.randint(100, 1000) for _ in car_names]
}

df = pd.DataFrame(data)
df
```

Out[83]:

	name	price	sell count
0	Nexon	22296	703
1	WagonR	16639	384
2	Swift	33024	928
3	Baleno	31049	990
4	Brezza	29628	106
5	Punch	24144	877
6	Creta	21717	925
7	Venue	20697	263
8	i20	42651	814
9	Seltos	17082	532
10	Sonet	16952	448
11	Thar	21140	384
12	Scorpio	29328	259
13	XUV700	30247	320
14	Alto	48118	881
15	Ertiga	16739	444
16	Fronx	28031	204
17	Dzire	42493	194
18	Harrier	29446	489
19	Tiago	44439	199

```
In [84]: carsdf = df.sort_values('sell count',ascending=False).head(6)
carsdf
# sorting dataframe based on sell count column in descending order
```

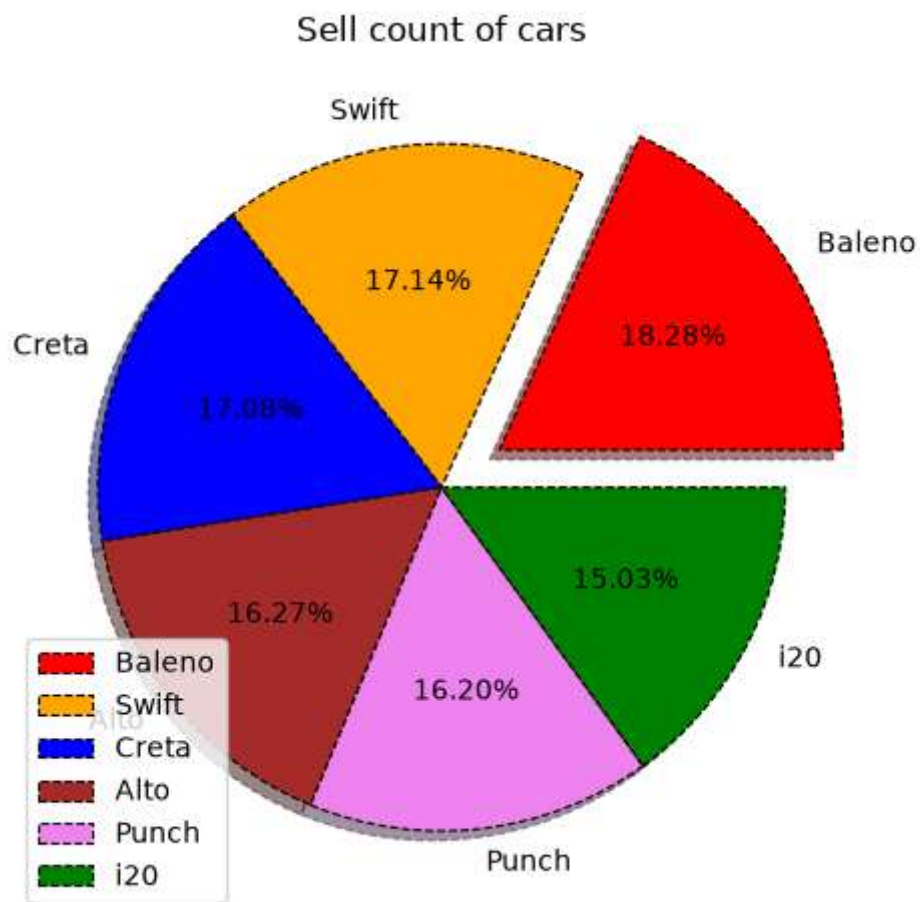
Out[84]:

	name	price	sell count
3	Baleno	31049	990
2	Swift	33024	928
6	Creta	21717	925
14	Alto	48118	881
5	Punch	24144	877
8	i20	42651	814

```
In [85]: plt.pie(carsdf['sell count'],
                labels=carsdf['name'],
                colors=['red', 'orange', 'blue', 'brown', 'violet', 'green'],
                shadow= 1,
                autopct='%1.2f%%',
                explode=[0.2,0,0,0,0,0],
                wedgeprops={'edgecolor':'black','linewidth':0.8, 'width':1.0,'linestyle':'-'})

plt.title('Sell count of cars')
plt.legend()
plt.tight_layout()
plt.show()

# Creates a pie chart showing market share for top 6 selling car models.
# Highlights the top seller with explode, custom colors, percentage labels, and das
```



In []:

4. Using Histogram show distribution of developers by their experience [use survey_data_sample.csv]

```
In [86]: df = pd.read_csv('survey_data_sample.csv')
df
```

Out[86]:

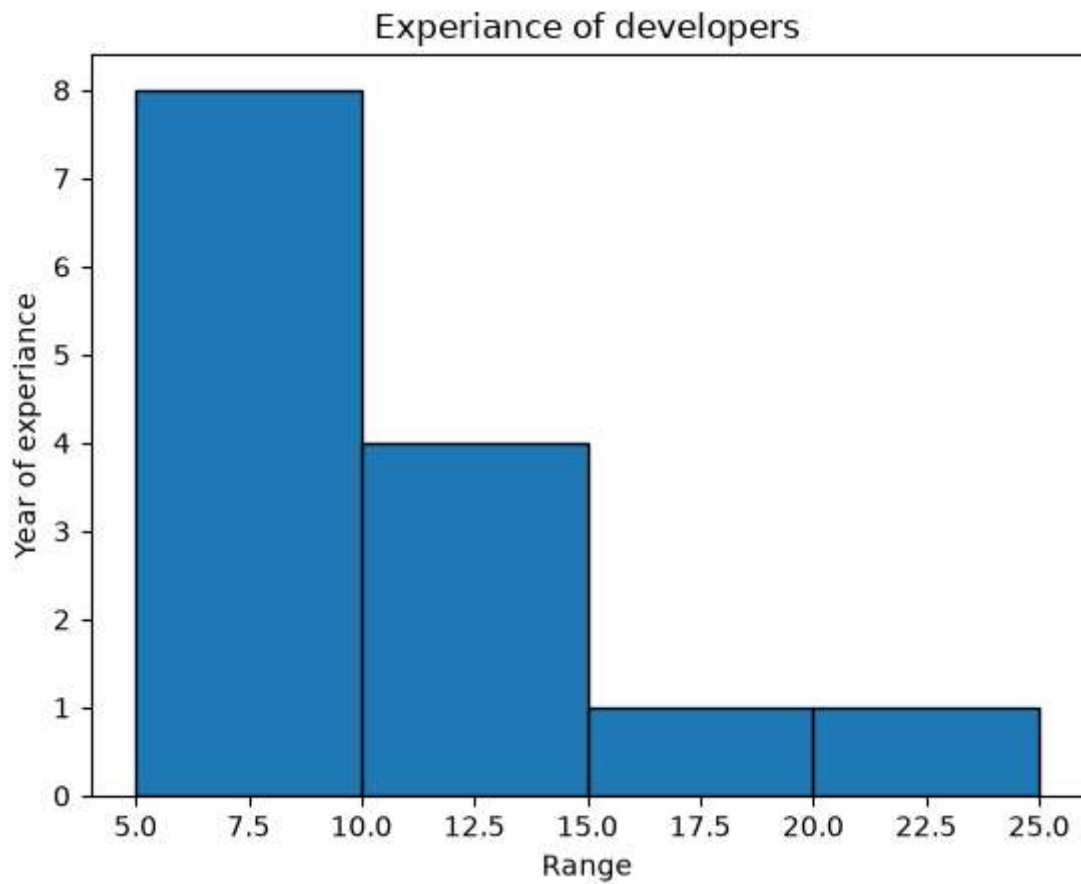
	RespondentID	DevType	YearsCodePro	Country	SalaryUSD
0	1	Full-stack developer	3	United States	75000
1	2	Back-end developer	5	India	18000
2	3	Front-end developer	1	Germany	48000
3	4	Full-stack developer	8	United States	110000
4	5	Data scientist	2	United Kingdom	52000
5	6	Mobile developer	4	India	15000
6	7	Back-end developer	12	Canada	95000
7	8	DevOps specialist	7	Germany	72000
8	9	Front-end developer	2	United States	65000
9	10	Full-stack developer	15	United Kingdom	125000
10	11	Back-end developer	6	India	22000
11	12	Data scientist	3	United States	88000
12	13	Full-stack developer	10	Germany	85000
13	14	Front-end developer	4	Canada	70000
14	15	Mobile developer	1	India	12000
15	16	Back-end developer	9	United States	115000
16	17	Full-stack developer	20	United Kingdom	140000
17	18	DevOps specialist	5	Germany	68000
18	19	Data scientist	8	United States	130000
19	20	Back-end developer	2	India	14000
20	21	Front-end developer	6	Canada	78000
21	22	Full-stack developer	11	United States	120000
22	23	Mobile developer	3	United Kingdom	58000
23	24	Back-end developer	14	Germany	92000
24	25	Full-stack developer	2	India	16000

```
In [87]: df['YearsCodePro']
bins = [5,10,15,20,25]
```

```
In [88]: plt.hist(df['YearsCodePro'], bins,edgecolor='black')
plt.title('Experiance of developers')
plt.xlabel('Range')
plt.ylabel('Year of experiance')
```

```
# Plots a histogram showing the distribution of professional coding experience among  
# Groups experience data into specified bin ranges with distinct black bar edges.
```

Out[88]: Text(0, 0.5, 'Year of experience')



In []:

In []:

In []: